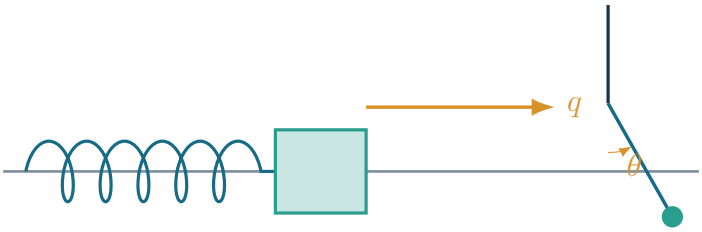
Euler–Lagrange Equations

Illustrated Problem Workbook

Fifteen progressive problems with spacious working pages and complete solutions



The habit this workbook develops

For every system: choose independent coordinates, express each position, calculate T and V, form $L = T - V$, and only then apply the Euler–Lagrange equations.

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Quick reference

For coordinates q_i,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0.}$$

For ordinary conservative systems,

$$L = T - V, \qquad p_i = \frac{\partial L}{\partial \dot{q}_i}.$$

If q_j is absent from L, then p_j is conserved. If L has no explicit time dependence and the kinetic energy is quadratic in velocities, then $T + V$ is conserved.

Before differentiating

State the coordinate direction, the zero of potential, and every geometric relation. Most sign errors occur before the Euler–Lagrange equation is used.

Progression

1. Problems 1–5: one coordinate and careful energy construction.
2. Problems 6–10: constraints, moving supports, cyclic coordinates, and effective potentials.
3. Problems 11–15: curved constraints, coupled systems, velocity-dependent terms, and normal modes.

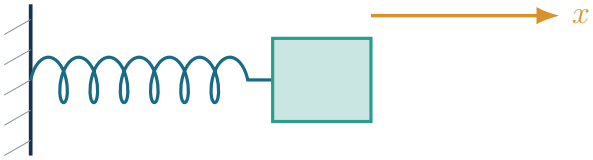
Problem 1: Harmonic oscillator

Focus: Applying the equation to a given Lagrangian

A particle has Lagrangian

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2.$$

1. Calculate $\partial L/\partial \dot{x}$ and $\partial L/\partial x$.
2. Derive the equation of motion.
3. Show that $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ is constant along a solution.
4. State the angular frequency.



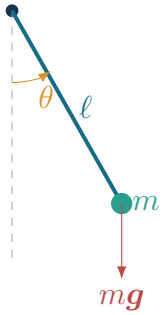
Working space

Problem 2: Simple pendulum

Focus: Building T and V from geometry

A bob of mass m is attached to a massless rod of length ℓ. Let θ be measured from the downward vertical.

1. Write the Cartesian position of the bob.
2. Show that $v^2 = \ell^2\dot{\theta}^2$.
3. Choose the lowest point as $V = 0$ and construct L.
4. Derive the exact equation of motion and its small-angle approximation.



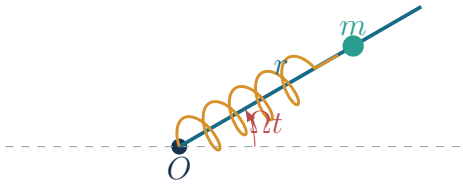
Working space

Problem 3: Bead on a rotating rod with a spring

Focus: Time-dependent geometry with one coordinate

A bead of mass m slides without friction along a horizontal rod rotating at constant angular speed Ω. A spring of stiffness k connects the bead to the rotation axis. Let r be the spring extension and take the natural length to be zero.

1. Show that $v^2 = \dot{r}^2 + \Omega^2 r^2$.
2. Construct the Lagrangian.
3. Derive the radial equation.
4. For what relation between k, m, and Ω is $r = 0$ stable?



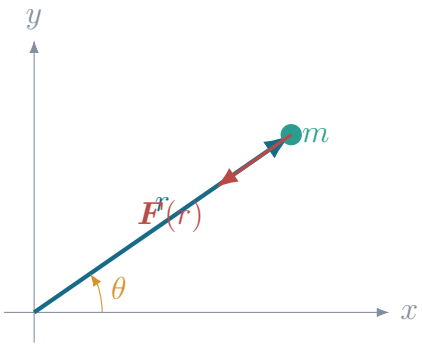
Working space

Problem 4: Particle in a central potential

Focus: Polar kinetic energy and a cyclic coordinate

A particle of mass m moves in a plane under a potential $V(r)$.

1. Use polar coordinates to write T and L.
2. Derive the r and θ equations.
3. Identify the conserved angular momentum J.
4. Eliminate $\dot{\theta}$ from the radial equation using J.



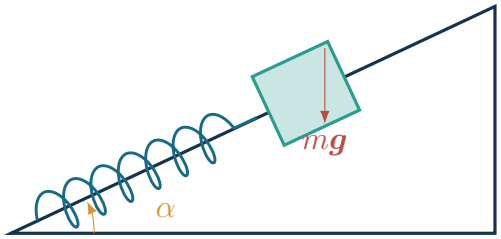
Working space

Problem 5: Spring on an inclined plane

Focus: Potential-energy signs and equilibrium

A block of mass m slides without friction down a plane inclined at angle α. A spring of stiffness k lies along the plane and has extension s, measured positively down the slope from its natural length.

1. Write T and the gravitational and spring potentials.
2. Derive the equation for $s(t)$.
3. Find the equilibrium extension.
4. Find the small-oscillation frequency about equilibrium.



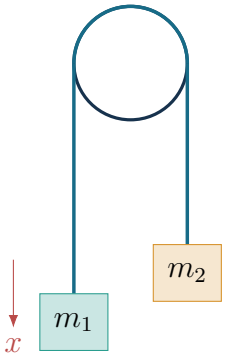
Working space

Problem 6: Atwood machine

Focus: Using a string constraint to remove a coordinate

Two masses $m_1 > m_2$ are joined by a light inextensible string over a frictionless pulley. Let x be the downward displacement of m_1; the upward displacement of m_2 is then also x.

1. Express the total kinetic and potential energies in terms of x.
2. Derive the acceleration.
3. Explain why the string tension does not appear in the reduced Lagrangian.



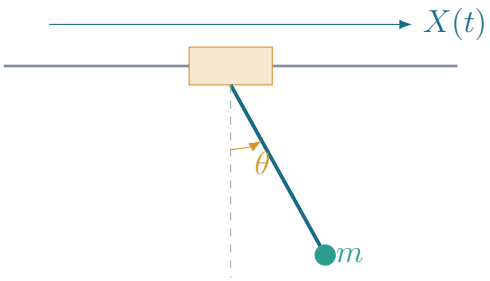
Working space

Problem 7: Pendulum with a moving support

Focus: An explicitly time-dependent Lagrangian

The support of a pendulum moves horizontally according to a prescribed function $X(t)$. The bob has mass m, the rod has length ℓ, and θ is measured from the downward vertical.

1. Write the bob coordinates x and y.
2. Construct T, V, and L.
3. Show that
$$\ell\ddot{\theta} + g\sin\theta + \ddot{X}\cos\theta = 0.$$
4. State the equation when $X(t) = A\cos\Omega t$.



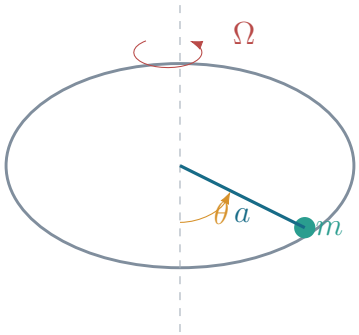
Working space

Problem 8: Bead on a rotating vertical hoop

Focus: Equilibria in a rotating constraint

A bead of mass m moves on a vertical circular hoop of radius a. The hoop rotates at constant angular speed Ω about its vertical diameter. Let θ be measured from the lowest point.

1. Show that
$$v^2 = a^2\dot{\theta}^2 + \Omega^2 a^2 \sin^2\theta.$$
2. Construct L and derive the equation of motion.
3. Find all equilibrium angles.
4. When do off-axis equilibria exist?



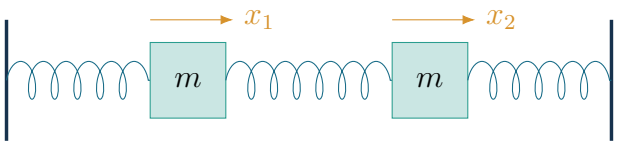
Working space

Problem 9: Two coupled oscillators

Focus: Coupled Euler–Lagrange equations and normal modes

Two equal masses m lie on a frictionless line. Three identical springs of stiffness k connect wall–mass–mass–wall. Let x_1, x_2 be displacements from equilibrium.

1. Construct T, V, and L.
2. Derive the two coupled equations.
3. Try $x_i = A_i \cos \omega t$ and find the two normal frequencies.
4. Describe the relative motion in each mode.



Working space

Problem 10: Kepler potential and effective energy

Focus: Cyclic coordinates and stability of circular motion

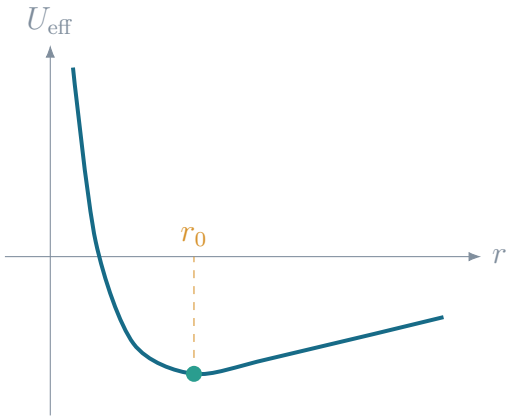
A particle moves in

$$V(r) = -\frac{\mu}{r}, \qquad \mu > 0.$$

1. Use the conserved angular momentum $J = mr^2\dot{\theta}$ to define

$$U_{\text{eff}}(r) = \frac{J^2}{2mr^2} - \frac{\mu}{r}.$$

2. Find the circular-orbit radius r_0.
3. Show that the circular orbit is stable by evaluating $U''_{\text{eff}}(r_0)$.
4. Find the angular frequency of the circular orbit.



Working space

Problem 11: Bead constrained to a parabola

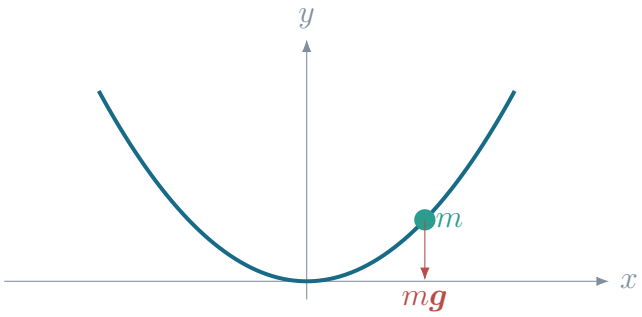
Focus: Kinetic energy on a curved track

A bead of mass m slides without friction on the vertical curve

$$y = ax^2, qquada > 0.$$

Use x as the coordinate and take gravity downward.

1. Show that
$$v^2 = (1 + 4a^2x^2)\dot{x}^2.$$
2. Construct the Lagrangian.
3. Derive the exact equation of motion.
4. Linearise near $x = 0$ and find the small-oscillation frequency.



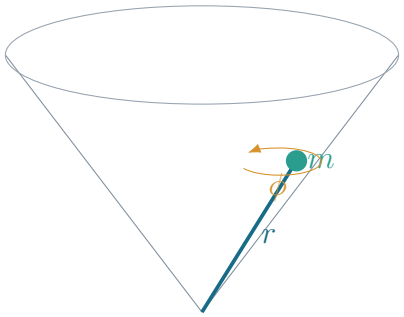
Working space

Problem 12: Bead on a cone

Focus: Two coordinates and an effective radial potential

A bead moves without friction on the cone $z = cr$, where (r, ϕ, z) are cylindrical coordinates and $c > 0$. Gravity acts downward.

1. Show that
$$T = \frac{1}{2}m\left[(1 + c^2)\dot{r}^2 + r^2\dot{\phi}^2\right].$$
2. Construct L and identify the conserved momentum.
3. Derive the radial equation.
4. Write the effective potential after eliminating $\dot{\phi}$.



Working space

Problem 13: Equal double pendulum

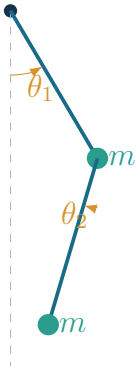
Focus: A genuinely coupled nonlinear system

Two equal masses m are joined by two massless rods of equal length ℓ. Angles θ_1, θ_2 are measured from the downward vertical.

1. Write both mass positions.
2. Show that, with $\Delta = \theta_1 - \theta_2$,

$$T = \frac{1}{2} m\ell^2 \left(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2 \cos\Delta\right).$$

3. Construct V and L.
4. Derive the two coupled Euler–Lagrange equations.



Working space

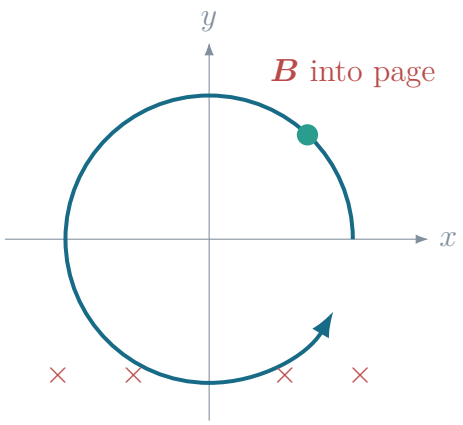
Problem 14: Charged particle in a magnetic field

Focus: A velocity-dependent Lagrangian

A particle of charge q and mass m moves in the xy-plane in a uniform magnetic field $B\boldsymbol{e}_z$. Consider

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{qB}{2}(x\dot{y} - y\dot{x}).$$

1. Derive the equations for x and y.
2. Show that the speed is constant.
3. Show that the motion is circular and find its angular frequency magnitude.
4. Explain why $T - V$ is too narrow a template for this example even though the Euler–Lagrange equations still apply.



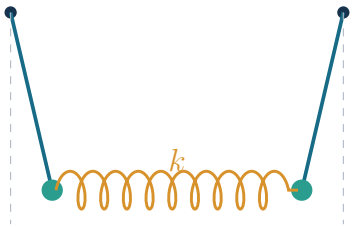
Working space

Problem 15: Coupled pendulums: small oscillations

Focus: Normal modes from a quadratic Lagrangian

Two equal pendulums of mass m and length ℓ are coupled by a weak horizontal spring of stiffness k joining the bobs. For small angles, the spring extension is approximately $\ell(\theta_2 - \theta_1)$.

1. Write the quadratic small-angle Lagrangian.
2. Derive the linear equations.
3. Find the in-phase and anti-phase normal frequencies.
4. Which mode stretches the coupling spring?



Working space

Worked solutions

Solution 1: Harmonic oscillator

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\partial L}{\partial x} = -kx.$$

Therefore

$$\boxed{m\ddot{x} + kx = 0.}$$

Multiply by $\dot{x}$:

$$m\ddot{x}\dot{x} + kx\dot{x} = 0 = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2\right).$$

Hence E is constant. Comparing with $\ddot{x} + \omega^2 x = 0$ gives

$$\boxed{\omega = \sqrt{k/m}.}$$

Solution 2: Simple pendulum

The bob position is

$$x = \ell \sin\theta, \qquad y = -\ell \cos\theta.$$

Thus

$$\dot{x} = \ell \cos\theta \dot{\theta}, \qquad \dot{y} = \ell \sin\theta \dot{\theta},$$

and $v^2 = \ell^2 \dot{\theta}^2$. With $V = mg\ell(1 - \cos\theta)$,

$$L = \frac{1}{2} m \ell^2 \dot{\theta}^2 - mg\ell(1 - \cos\theta).$$

The equation is

$$\boxed{\ddot{\theta} + \frac{g}{\ell} \sin\theta = 0.}$$

For small angles, $\sin\theta \approx \theta$:

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0.$$

Solution 3: Bead on a rotating rod with a spring

In polar coordinates with prescribed $\theta = \Omega t$,

$$v^2 = \dot{r}^2 + r^2\Omega^2.$$

Therefore

$$L = \frac{1}{2}m(\dot{r}^2 + \Omega^2 r^2) - \frac{1}{2}kr^2.$$

Now

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r}, \qquad \frac{\partial L}{\partial r} = (m\Omega^2 - k)r.$$

Hence

$$\boxed{m\ddot{r} + (k - m\Omega^2)r = 0.}$$

The origin is stable when

$$\boxed{k > m\Omega^2.}$$

At equality the quadratic restoring term vanishes.

Solution 4: Particle in a central potential

The polar speed is

$$v^2 = \dot{r}^2 + r^2\dot{\theta}^2,$$

so

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r).$$

For r,

$$\boxed{m\ddot{r} - mr\dot{\theta}^2 + V'(r) = 0.}$$

For θ,

$$\frac{\mathrm{d}}{\mathrm{d}t}(mr^2\dot{\theta}) = 0, \qquad \boxed{J = mr^2\dot{\theta}.}$$

Substitution gives

$$\boxed{m\ddot{r} - \frac{J^2}{mr^3} + V'(r) = 0.}$$

Solution 5: Spring on an inclined plane

With s positive down the slope,

$$T = \frac{1}{2}m\dot{s}^2, \qquad V = \frac{1}{2}ks^2 - mgs\sin\alpha.$$

Thus

$$L = \frac{1}{2}m\dot{s}^2 - \frac{1}{2}ks^2 + mgs\sin\alpha.$$

The equation is

$$\boxed{m\ddot{s} + ks - mg\sin\alpha = 0.}$$

At equilibrium,

$$\boxed{s_{\text{eq}} = \frac{mg\sin\alpha}{k}.}$$

Writing $s = s_{\text{eq}} + \xi$ gives $m\ddot{\xi} + k\xi = 0$, so $\omega = \sqrt{k/m}$.

Solution 6: Atwood machine

The string constraint gives both masses the same speed magnitude:

$$T = \frac{1}{2}(m_1 + m_2)\dot{x}^2.$$

As m_1 moves down by x, its potential decreases by $m_1 g x$; as m_2 moves up, its potential increases by $m_2 g x$:

$$V = (m_2 - m_1)gx.$$

Therefore

$$L = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + (m_1 - m_2)gx.$$

The equation is

$$(m_1 + m_2)\ddot{x} = (m_1 - m_2)g,$$

so

$$\boxed{\ddot{x} = \frac{m_1 - m_2}{m_1 + m_2}g.}$$

The internal tension is absent because the constraint has already reduced the system to its allowed motion.

Solution 7: Pendulum with a moving support

The bob position is

$$x = X(t) + \ell \sin\theta, \qquad y = -\ell \cos\theta.$$

Hence

$$T = \frac{1}{2} m \left(\dot{X}^2 + 2\ell \dot{X} \cos\theta \dot{\theta} + \ell^2 \dot{\theta}^2 \right), \qquad V = -mg\ell \cos\theta.$$

After differentiating, the terms proportional to $\dot{X}\dot{\theta}\sin\theta$ cancel, leaving

$$\boxed{\ell\ddot{\theta} + g\sin\theta + \ddot{X}\cos\theta = 0.}$$

For $X = A\cos\Omega t$,

$$\ell\ddot{\theta} + g\sin\theta - A\Omega^2 \cos\Omega t \cos\theta = 0.$$

Solution 8: Bead on a rotating vertical hoop

The distance from the rotation axis is $a\sin\theta$, while motion along the hoop contributes $a\dot{\theta}$. Thus

$$T = \frac{1}{2}ma^2\left(\dot{\theta}^2 + \Omega^2\sin^2\theta\right), \qquad V = -mga\cos\theta.$$

Therefore

$$L = \frac{1}{2}ma^2(\dot{\theta}^2 + \Omega^2\sin^2\theta) + mga\cos\theta.$$

The equation is

$$\boxed{\ddot{\theta} = \Omega^2\sin\theta\cos\theta - \frac{g}{a}\sin\theta.}$$

Equilibria satisfy

$$\sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right) = 0.$$

Thus $\theta = 0, \pi$, and off-axis equilibria obey

$$\boxed{\cos\theta = \frac{g}{a\Omega^2}.}$$

They exist when $\Omega^2 \geq g/a$.

Solution 9: Two coupled oscillators

The energies are

$$T = \frac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2),$$

$$V = \frac{1}{2}kx_1^2 + \frac{1}{2}k(x_2 - x_1)^2 + \frac{1}{2}kx_2^2.$$

The equations are

$$\boxed{m\ddot{x}_1 + 2kx_1 - kx_2 = 0,} \qquad \boxed{m\ddot{x}_2 + 2kx_2 - kx_1 = 0.}$$

For $(A_1, A_2) = (1, 1)$,

$$\omega_+^2 = \frac{k}{m};$$

the middle spring is unstretched. For $(1, -1)$,

$$\omega_-^2 = \frac{3k}{m};$$

the masses move oppositely and the middle spring stretches most strongly.

Solution 10: Kepler potential and effective energy

The radial energy is

$$E = \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r), \qquad U_{\text{eff}} = \frac{J^2}{2mr^2} - \frac{\mu}{r}.$$

A circular orbit is a stationary point:

$$U'_{\text{eff}} = -\frac{J^2}{mr^3} + \frac{\mu}{r^2} = 0.$$

Hence

$$\boxed{r_0 = \frac{J^2}{m\mu}.}$$

Also

$$U''_{\text{eff}}(r_0) = \frac{3J^2}{mr_0^4} - \frac{2\mu}{r_0^3} = \frac{\mu}{r_0^3} > 0,$$

so the orbit is stable. Using $mr_0\omega^2 = \mu/r_0^2$,

$$\boxed{\omega^2 = \frac{\mu}{mr_0^3}.}$$

Solution 11: Bead constrained to a parabola

Since $y = ax^2$,

$$\dot{y} = 2ax\dot{x}, \qquad v^2 = \dot{x}^2 + \dot{y}^2 = (1 + 4a^2x^2)\dot{x}^2.$$

Thus

$$L = \frac{1}{2}m(1 + 4a^2x^2)\dot{x}^2 - mgax^2.$$

The exact equation is

$$\boxed{(1 + 4a^2x^2)\ddot{x} + 4a^2x\dot{x}^2 + 2gax = 0.}$$

Near $x = 0$, nonlinear terms are negligible:

$$\ddot{x} + 2ga\,x = 0, \qquad \boxed{\omega = \sqrt{2ga}.}$$

Solution 12: Bead on a cone

With $z = cr$, $\dot{z} = c\dot{r}$, so

$$T = \frac{1}{2}m[(1+c^2)\dot{r}^2 + r^2\dot{\phi}^2], \qquad V = mgcr.$$

The angle is cyclic:

$$\boxed{J = mr^2\dot{\phi}.}$$

The radial equation is

$$\boxed{m(1+c^2)\ddot{r} - mr\dot{\phi}^2 + mgc = 0.}$$

Eliminating $\dot{\phi}$ gives

$$\frac{1}{2}m(1+c^2)\dot{r}^2 + U_{\text{eff}}(r) = E,$$

with

$$\boxed{U_{\text{eff}}(r) = \frac{J^2}{2mr^2} + mgcr.}$$

Solution 13: Equal double pendulum

The positions are

$$x_1 = \ell \sin\theta_1, \quad y_1 = -\ell\cos\theta_1,$$

$$x_2 = x_1 + \ell\sin\theta_2, \quad y_2 = y_1 - \ell\cos\theta_2.$$

This yields

$$T = \frac{1}{2}m\ell^2(2\dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2\cos\Delta),$$

$$V = -mg\ell(2\cos\theta_1 + \cos\theta_2).$$

The equations are

$$\boxed{2\ddot\theta_1 + \ddot\theta_2\cos\Delta + \dot\theta_2^2\sin\Delta + 2\frac{g}{\ell}\sin\theta_1 = 0,}$$

$$\boxed{\ddot\theta_2 + \ddot\theta_1\cos\Delta - \dot\theta_1^2\sin\Delta + \frac{g}{\ell}\sin\theta_2 = 0.}$$

Solution 14: Charged particle in a magnetic field

For x,

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} - \frac{qB}{2}y, \qquad \frac{\partial L}{\partial x} = \frac{qB}{2}\dot{y}.$$

For y the analogous calculation gives

$$\boxed{m\ddot{x} = qB\dot{y},} \qquad \boxed{m\ddot{y} = -qB\dot{x}.}$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}(\dot{x}^2 + \dot{y}^2) = 2\dot{x}\ddot{x} + 2\dot{y}\ddot{y} = 0,$$

so the speed is constant. The acceleration is always perpendicular to velocity, producing circular motion with

$$\boxed{|\omega_c| = \frac{|qB|}{m}.}$$

The term linear in velocity represents magnetic coupling through a vector potential; the Euler–Lagrange formalism is broader than the simple template $T - V$.

Solution 15: Coupled pendulums: small oscillations

To quadratic order,

$$T = \frac{1}{2} m\ell^2(\dot\theta_1^2 + \dot\theta_2^2),$$

$$V = \frac{1}{2} mg\ell(\theta_1^2 + \theta_2^2) + \frac{1}{2} k\ell^2(\theta_2 - \theta_1)^2.$$

The equations are

$$m\ell^2\ddot\theta_1 + mg\ell\theta_1 + k\ell^2(\theta_1 - \theta_2) = 0,$$

$$m\ell^2\ddot\theta_2 + mg\ell\theta_2 + k\ell^2(\theta_2 - \theta_1) = 0.$$

For the in-phase mode $(1, 1)$, the spring does not stretch:

$$\boxed{\omega_{\text{in}}^2 = \frac{g}{\ell}.}$$

For the anti-phase mode $(1, -1)$,

$$\boxed{\omega_{\text{anti}}^2 = \frac{g}{\ell} + \frac{2k}{m}.}$$

Only the anti-phase mode stretches the coupling spring to first order.

Final checklist

Before accepting an equation of motion, ask:

- Are the coordinates independent?
- Does every term in T have units of energy?
- Is the zero and sign of V declared?
- Does a cyclic coordinate give the expected conserved quantity?
- Does the equation reduce correctly when a coupling, rotation, or angle is set to zero?
- Is the small-angle or small-displacement approximation stated only after the exact equation?

What fluency looks like

The goal is not to memorise fifteen answers. It is to make the sequence

$$\text{geometry} \to \text{coordinates} \to T, V \to L \to \text{equations}$$

feel routine enough that attention can move to interpretation.